

A Study: Speed Stat Ceiling Breaks

```
In [2]: import pandas as pd
import os
import numpy as np
import bs4
import seaborn as sns
import matplotlib.pyplot as plt
```

Finding All Ceiling Breaks

Let's find all ceiling breaks, who did it, when, and what version.

```
In [3]: table = pd.read_csv('../data/lv40_table.csv')
date = table.sort_values(by='release')
breaks = pd.DataFrame(columns=['hero', 'spd', 'release', 'version'])
```

```
In [4]: fastest = date[date['release'] == date.loc[0]['release']].sort_values(by='spd').iloc[-1]['spd']
breaks.loc[0] = [date[date['release'] == date.loc[0]['release']].sort_values(by='spd').iloc[-1]['hero'],
                fastest, date.loc[0]['release'], date.loc[0]['version']]
for i in range(date.shape[0]):
    row = date.iloc[i]
    if row['spd'] > fastest:
        fastest = row['spd']
        breaks.loc[breaks.shape[0]] = [row['hero'], fastest, row['release'], row['version']]
```

```
In [5]: breaks
```

```
Out[5]:
```

	hero	spd	release	version
0	Lon'qu: Solitary Blade	39	2017-02-02	1.00
1	Mia: Lady of Blades	40	2017-11-15	1.80
2	Mareeta: The Blade's Pawn	41	2019-05-10	3.50
3	Mareeta: Sword of Stars	42	2019-12-06	4.00
4	Freyja: Lady of Nightmare	44	2020-11-27	4.11
5	Corrin: Starry Seer	45	2022-10-06	6.10
6	Ursula: Royal-Blue Crow	46	2023-01-10	7.10
7	Karla: Spring Reveries	47	2023-03-07	7.30
8	Bernadetta: Sheltered Loner	48	2024-08-16	8.80
9	Sharena: Hopeful Princess	49	2025-08-15	9.80

Analysis

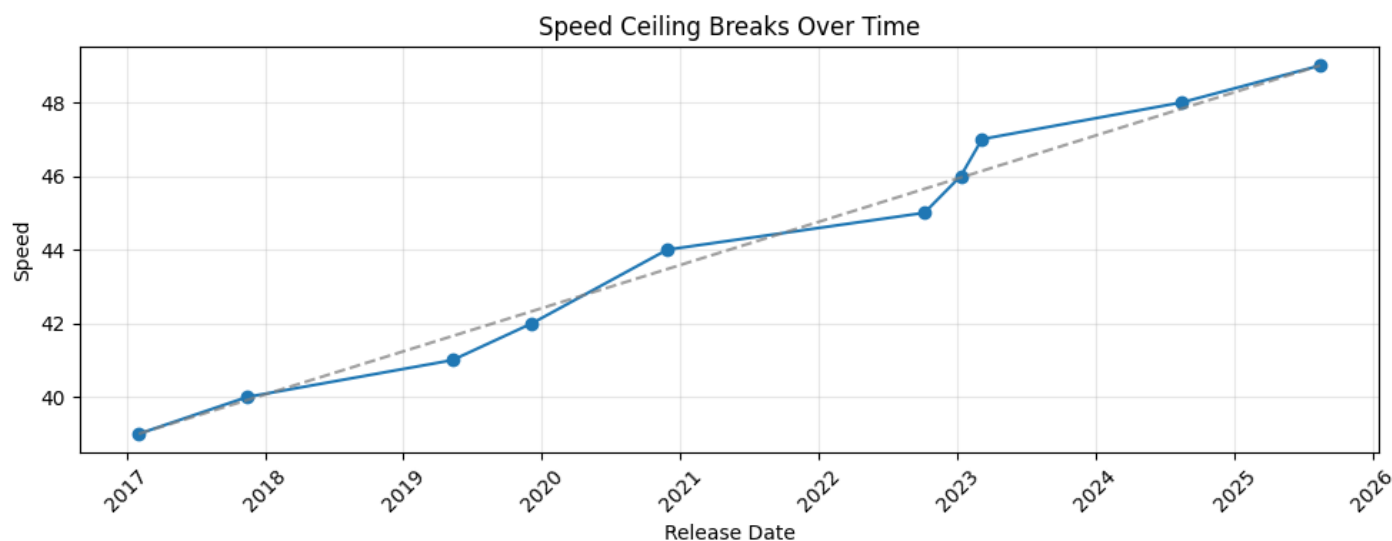
Release Dates

So there's no guarantee that a year or major version will see a speed ceiling break.

Also, there doesn't seem to be a clear pattern to the time between ceiling breaks.

```
In [6]: breaks["release"] = pd.to_datetime(breaks["release"])

plt.figure(figsize=(10, 4))
plt.plot(breaks["release"], breaks["spd"], marker="o")
plt.plot([breaks["release"].iloc[0], breaks["release"].iloc[-1]],
         [breaks["spd"].iloc[0], breaks["spd"].iloc[-1]],
         linestyle='--', alpha=0.75, color='gray')
plt.title("Speed Ceiling Breaks Over Time")
plt.xlabel("Release Date")
plt.ylabel("Speed")
plt.xticks(rotation=45)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



(Note that the slopes of each segment do not signify that the speed stat was slowly increasing over time for units. This is simply to visualize the speed at which the next ceiling was broken.)

Something that I immediately noticed was that **43 speed was completely skipped over?? Freyja: Lady of Nightmare** (Mythic) went straight to 44 speed when she released.

So who was the first 43 speed unit, and when did they appear?

```
In [7]: first43 = np.sort(date[date['spd'] == 43]['release'].unique())[0]
first43
```

```
Out[7]: '2021-01-18'
```

```
In [8]: spd43 = date[date['release'] == first43]
spd43 = spd43[spd43['spd'] == 43]
spd43['hero']
```

```
Out[8]: 639    Katarina: Torchlit Wanderer
Name: hero, dtype: object
```

The first 43 speed hero, which was **Katarina: Torchlit Wanderer** appeared around two months after the first 44 speed unit.

What about significance? Is there any significance to the versions or release dates of ceiling-breakers?

Ursula: Royal-Blue Crow was a unit on the yearly desert-themed special banner in January, specifically for the 2023 Khadein banner (and so was **Katarina: Torchlit Wanderer**, who was on the 2022 Plegian banner).

The last two breaks were done by **Sharena: Hopeful Princess** and **Bernadetta: Sheltered Loner**. Both are significant in that they are both Choose Your Legends units. Specifically, they were the first place winners of the female division for CYL 9 and CYL 8, respectively. Could we say that the next speed break will happen with the release of the CYL 10 units? Maybe. It'd be interesting.

Weapon Type

Let's first get the rest of the data for the units in `breaks`.

```
In [9]: breaks = breaks[['hero']].merge(table, on='hero')
breaks.head()
```

Out[9]:

	hero	entry	move	weapon	hp	atk	spd	def	res	total	...	tempest	legendary	entwii
0	Lon'qu: Solitary Blade	Awakening	Infantry	Red Sword	45	29	39	22	22	157	...	False	False	Fi
1	Mia: Lady of Blades	Path of Radiance	Infantry	Red Sword	38	32	40	28	25	163	...	False	False	Fi
2	Mareeta: The Blade's Pawn	Thracia 776	Infantry	Red Sword	42	33	41	29	23	168	...	False	False	Fi
3	Mareeta: Sword of Stars	Thracia 776	Infantry	Red Sword	41	35	42	31	23	172	...	False	False	Fi
4	Freyja: Lady of Nightmare	Heroes	Cavalry	Green Beast	40	38	44	24	22	168	...	False	False	Fi

5 rows × 31 columns

What does the distribution look across color? Across weapon types?

```
In [10]: breaks['color'].value_counts()
```

```
Out[10]: color
Red      4
Green    4
Blue     2
Name: count, dtype: int64
```

No Colorless representation, unfortunately.

```
In [11]: breaks['weapon'].value_counts()
```

```
Out[11]: weapon
Red Sword      4
Green Tome     2
Green Beast    1
Blue Breath    1
Green Axe      1
Blue Bow       1
Name: count, dtype: int64
```

Pretty on point, I suppose? For god infantry units, Godswords are a favored build. Even after **Saber: Dune Walker** dropped with his **Jehanna Lance**, I saw Godswords more often than Godlances.

But, an analysis on the most common infantry archetypes is something for another time. Maybe I'll link it here when I get to it one day.

Hero Availability

Is there any significance to how accessible these heroes are?

```
In [13]: def bettervis(cond):
        if cond:
            return 'O'
        return ' '

temp = breaks[['hero', 'regular_4_3_2_1', 'specialRate', 'regular_5', 'mythic', 'special']].copy()
for i in temp.columns[1:]:
    temp[i] = temp[i].apply(bettervis)
temp
```

Out[13]:

	hero	regular_4_3_2_1	specialRate	regular_5	mythic	special
0	Lon'qu: Solitary Blade	O				
1	Mia: Lady of Blades		O			
2	Mareeta: The Blade's Pawn		O			
3	Mareeta: Sword of Stars		O			
4	Freyja: Lady of Nightmare				O	
5	Corrin: Starry Seer		O			O
6	Ursula: Royal-Blue Crow		O			O
7	Karla: Spring Reveries					O
8	Bernadetta: Sheltered Loner			O		
9	Sharena: Hopeful Princess			O		

Breaks started with only normal pool units. After **Freyja: Lady of Nightmare** breaks, the following breakers are all seasonals (incentive for players to pull limited heroes?). At least, until Choose Your Legends units become breakers. Nothing too significant? Their reasoning for certain units being breakers is up for speculation. There is incentive to pull limited units because of their limited availability and likely new skills. Same goes for CYL units that always bring something new to the table in terms of skills and effects.

More coming soon...